

SWATHI JAYAKUMAR

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PROFILE SUMMARY

Strong foundation in programming with C++ and C, and proficient in Java and Python. Experienced in full-stack development, with hands-on expertise in machine learning, deep learning, computer vision, and LLM-powered solutions. Experienced in database design, API testing, and automation frameworks. Passionate about technology, driven to solve real-world problems, and committed to building impactful, scalable solutions in computer science.

EDUCATION

Amrita Vishwa Vidyapeetham, Bachelor of Technology in Computer Science and Engineering 2022 – Present

- CGPA: 8.32 / 10

- Campus: Kollam, Kerala

Cochin Refineries School, Senior Secondary (Class 12) 2019 – 2021

- Percentage: 95.2%

- Location: Ernakulam, Kerala

Cochin Refineries School, Secondary (Class 10) 2017 – 2019

- Percentage: 88.6%

- Location: Ernakulam, Kerala

EXPERIENCE

Intern, Founding Minds Software – Infopark, Kochi Aug 2024

- Participated in the **ICPR 2024 Multi-Modal Visual Pattern Recognition Challenge - Track 2: Detection**
- Developed object detection models for multi-modal data streams (RGB, Depth, Infrared)

PUBLICATIONS

Advancing Multimodal Visual Pattern Recognition: Object Detection June 2025

- Published in the Springer proceedings of the **International Conference on Pattern Recognition (ICPR) Workshop**

PROJECTS

AgroConnect – Farmer to Market Direct Trade Platform May 2025 – June 2025

- Designed and developed a secure full-stack web application enabling farmers to list produce and buyers to place direct orders, improving trading efficiency and eliminating middlemen
- Implemented image upload, cart management, and authentication features to enhance user experience and platform security
- **Tech:** Spring Boot, Angular 17, MySQL, JWT, REST APIs

One-View Amazon Product Analyzer April 2025

- Built a Python-based application to extract and analyze Amazon product reviews, performing sentiment classification and review summarization for better buyer decision-making
- Integrated real-time question-answering functionality using NLP to allow users to interact with summarized feedback
- **Tech:** Python, BeautifulSoup, NLTK, Flask

Demand Forecasting for Walmart November 2024

- Created a Random Forest model to forecast daily Walmart sales using time-series data, achieving 95.6% prediction accuracy
- Delivered actionable insights for inventory management and supply chain optimization through feature

engineering and hyperparameter tuning

- **Tech:** Python, Pandas, Scikit-learn, Matplotlib, Seaborn

Multi-Modal Detection System

August 2024

- Designed and implemented an object detection pipeline using YOLOv8 for multimodal input (RGB, Depth, Infrared), improving detection robustness in varied environments
 - Optimized model performance using Ray Tune, enabling real-time processing capabilities
 - **Tech:** YOLOv8, PyTorch, OpenCV
- enumitem

Technical Skills

- **Languages:** Python, Java, C++ , C.
- **Web Development:** HTML, CSS, JavaScript, TypeScript, Angular, Spring Boot.
- **Databases:** PostgreSQL, MySQL.
- **Testing & Tools:** GitHub, Postman (API Testing).
- **Machine Learning & AI:** PyTorch, TensorFlow, Hugging Face, LangChain.
- **Soft Skills:** Problem-Solving, Communication, Team Collaboration, Leadership, Adaptability

COURSES/CERTIFICATIONS

Machine Learning with Python – IBM

Completed

- Gained practical knowledge of machine learning techniques including regression, classification, clustering, and model evaluation using Python
- **Tools Used:** Python, scikit-learn, Pandas, Jupyter Notebooks

WORKSHOPS

Tredence Hackathon 2025 Finalist

March 2025

- Selected among Top 11 teams nationwide
- Built a SaaS platform for multimodal AI customer support using LLMs and speech models
- Focused on real-time contextual conversations with enterprise support use cases

SOCIAL OUTREACH

Drug Abuse Awareness Session – NSS Higher Secondary School, Alappuzha

October 2024

- Conducted a session on drug abuse awareness as part of the Socially Responsible Student (SSR) program
- Educated students on harmful impacts of substance abuse and promoted prevention strategies